What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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Transaction-level route structures

Direct swap

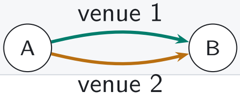
one transaction

Sequential vehicle route

one transaction

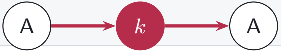
Parallel split

one transaction



Round trip

one transaction



Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Measuring vehicle dominance

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.

$$i \longrightarrow k \longrightarrow o$$

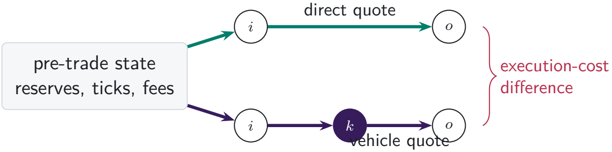
Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

The blockchain reveals chosen routes and contemporaneous alternatives

RESEARCH DESIGN

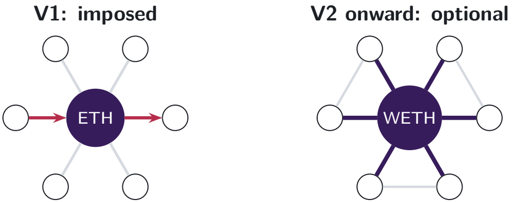
- FX data typically reveal executed trades.
- On-chain reserves and fees also let us price feasible alternatives.
- The comparison holds market conditions fixed at execution.



Protocol design first imposed ETH intermediation

INSTITUTIONAL FACT

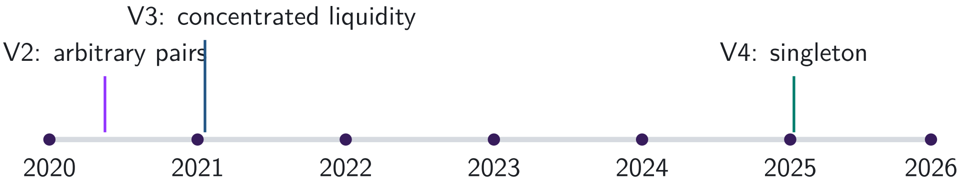
- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



ETH remained central after arbitrary pairs became possible.

The panel follows the market from V1 through V4

DATA



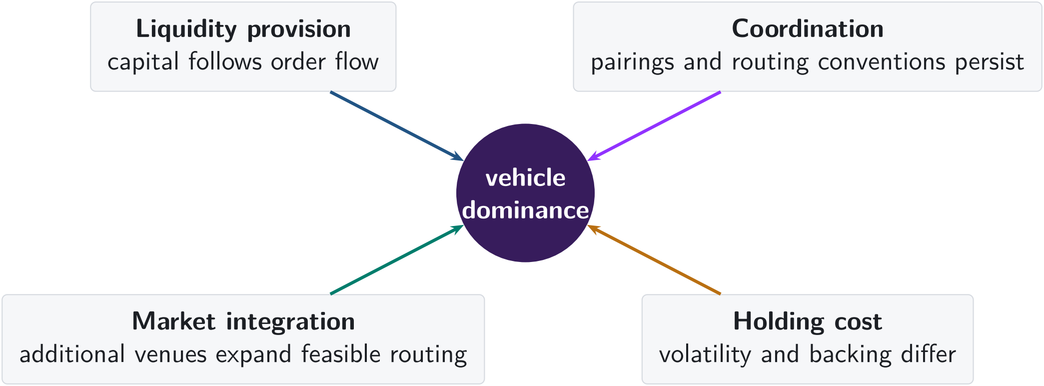
Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Four measures capture use, scale, and network position

1. **Vehicle share**
   Fraction of route units intermediated by asset k.
2. **Excess use**
   Intermediary share relative to endpoint demand.
3. **Network position**
   Degree, strength, and betweenness on the trading graph.
4. **Execution advantage**
   Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Together, the measures distinguish use, scale, network position, and execution cost.

Four forces can sustain vehicle dominance



The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes
Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation
Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

Contemporaneous alternatives
An incumbent retains volume despite a cheaper feasible route.

how the role begins

how its share moves

how long it survives

Architecture exposure changes within cells

PROVISIONAL: LIVE SCIENCE

Below threshold $s_{iok,t}^{V4} < c$ $\xrightarrow{\text{entry}}$ **V4-active state** $s_{iok,t}^{V4} \geq c$ $\xrightarrow{\text{exit}}$ **Below threshold** $s_{iok,t}^{V4} < c$

State changes require three consecutive weeks and are detected within the same ordered endpoint pair and candidate vehicle. Thresholds $c \in \{5\%, 10\%, 25\%\}$.

Measured architecture exposure

- realised V4 share in the cell
- may enter, exit, or re-enter

Outcome

- overall V3+V4 vehicle-route share
- adjusted within endpoint-pair × week

Calendar date is not treatment. Exposure is endogenous at E0; re-entry is a new entry after exit.

Entry, exit, and persistence are distinct objects

PROVISIONAL: LIVE SCIENCE

Observed pattern	Reading	Identification warning
Rise on entry; fall on exit	Association consistent with an architecture channel	Adoption can still respond to expected demand
Rise on entry; persists after exit	Persistence in vehicle use	Hysteresis needs asymmetric retention versus displacement under cost-state reversal
Movement before entry	Anticipation or endogenous adoption	A launch-date design would misattribute movement
No response to state changes	Architecture is not the operative margin	Report support and power before reading the null

Exit reverses measured architecture exposure, not necessarily the vehicle role. Primary contrasts use isolated ± 8-week windows; contaminated events remain visible in the support audit.

Live architecture audit: what the current run must establish

PROVISIONAL: LIVE SCIENCE

	5%	10%	25%
Sustained entries	awaiting Studio-current certified generation		
Substitution exits	awaiting Studio-current certified generation		
Clean isolated windows	to be established from certified support		
Entry contrast	to be estimated with pair-week adjustment		
Exit contrast	to be estimated with pair-week adjustment		

FEEDBACK NOW

- Is three-week confirmation economically sensible?
- Which threshold should be primary?
- Should exit include disappearance or only architecture substitution?
- What separates persistence from delayed adjustment?

No coefficient enters the paper from this slide. Each populated result will name its exact Studio data generation.

Three questions organize the evidence

1. **Formation**
   When does market architecture create or strengthen an intermediary role?
2. **Rotation**
   Which changes reflect intermediary demand, endpoint demand, trade size, or routing technology?
3. **Persistence**
   How long does incumbent intermediation survive when contemporaneous alternatives are available?

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

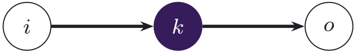
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Backing regimes vary over time

Native platform asset
ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised
DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

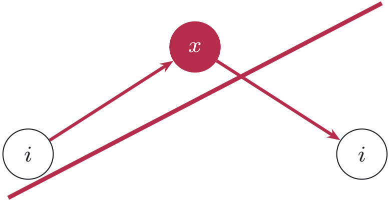
Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

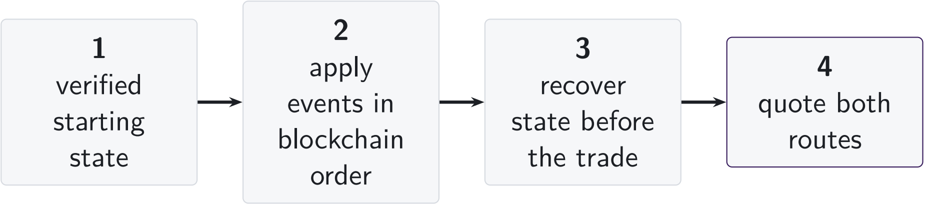
A3. Sample restrictions preserve value measurement

- Round trips are excluded from intermediation measures.
- Dollar values require consistent prices and verified token identities.
- We retain routes whose two legs imply dollar values within the pre-specified tolerance.



round trip: excluded from value-weighted vehicle measures

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. AMM designs require different state variables

Constant product Reserves, fee, and invariant.

Concentrated liquidity Active liquidity, price, tick, and initialised-tick changes.

StableSwap families Balances, amplification, rates, and oracle state.

Weighted pools Vault balances, weights, scaling, and rate providers.

V4 singleton Pool key, tick state, and fee for standard pools without custom hooks.

Estimates include protocol families whose balances, fees, and pricing state can be independently validated.

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.

same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions

1. **Primary panel** day observations; date fixed effects
2. **Calendar sensitivity** day observations; week and weekday fixed effects
3. **Aggregation robustness** complete weeks; week fixed effects
4. **Transition series** exact 1-, 7-, 30-, and 120-day links; HAC inference

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

A8. Count and value define separate dominance margins

Count-weighted share

$$S^{N}_{k,t} = \frac{N^{I}_{k,t}}{\sum_j N^{I}_{j,t}}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S^{V}_{k,t} = \frac{\mathrm{Vol}^{I}_{k,t}}{\sum_j \mathrm{Vol}^{I}_{j,t}}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S^{I}_{k,t} = \frac{N^{I}_{k,t}}{\sum_j N^{I}_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$S^{E}_{k,t} = \frac{N^{\mathrm{in}}_{k,t} + N^{\mathrm{out}}_{k,t}}{\sum_j N^{\mathrm{in}}_{j,t} + N^{\mathrm{out}}_{j,t}}$$

How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are distinct quantities

Deposited capital
Funds committed by liquidity providers, established from independently valued holdings or audited claims. **Inventory**

Exact token amounts held by the pool.

Executable depth
Amount that can be traded for a given price impact under current pool state.
Quote quality

Execution price for trade size q along feasible routes.

Capital enters only where holdings and ownership are independently validated; execution depth comes from the pool's pricing function.

A11. Integration changes both reach and route choice

Available trading opportunities

- active venues
- feasible paths
- available intermediary assets

Route choice conditional on availability

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

We test whether stablecoin intermediation rises more when additional venues and routes are available, conditional on endpoint pair and date.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage